Tail Risk Estimation under Heavy Tailed Distributions: A Monte Carlo Study

Dipesh Lamichhane
dipeshkiran012@gmail.com

Abstract

Accurate estimation of extreme tail risk measures is central to sound financial decision making under uncertainty. This study examines Monte Carlo estimators of Value at Risk (VaR) and Expected Shortfall (ES) under realistic features of financial returns, including heavy tails, finite sample uncertainty, model misspecification, and time varying volatility. We analyze quantile estimators through the lens of the quantile central limit theorem and finite sample concentration inequalities based on the Dvoretzky Kiefer Wolfowitz (DKW) inequality. The asymptotic confidence interval employs a Gaussian kernel density estimator with Silverman's rule of thumb bandwidth to approximate the tail density. Variance reduction strategies, antithetic variates and Gaussian mean shift importance sampling directed along the portfolio loss direction, are evaluated for their effectiveness in tail regions. Model risk is assessed by generating synthetic return paths from a multivariate Student t distribution and forecasting with both Gaussian and Student t models. Dynamic volatility is captured via univariate GARCH(1,1) models fit to portfolio returns, with Gaussian and Student t innovations, using returns scaled by 100 for numerical stability and the model refitted every 20 observations. Rolling backtests over 1547 test observations demonstrate that static parametric models tend to underestimate tail risk at the 99% level, leading to elevated exception rates relative to nominal targets. GARCH models with Student t innovations provide substantially improved unconditional and conditional coverage, confirmed by Kupiec and Christoffersen tests and the Fissler Ziegel proper scoring rule. These findings underscore the sensitivity of tail risk estimation to distributional assumptions under heavy tailed data and motivate the development of robust risk measurement frameworks.

1 Introduction

Decision making under uncertainty is a foundational challenge in finance, insurance, and operations research. Tail risk measures, quantities that characterize the behavior of losses in the extreme right tail of the loss distribution, are used to set regulatory capital requirements, determine position limits, and stress test portfolios. Two measures have achieved canonical status: Value at Risk (VaR) and Expected Shortfall (ES, also known as Conditional Value at Risk) [2].

Standard practice often assumes that portfolio returns follow a multivariate Gaussian distribution. This is analytically convenient but empirically problematic: financial returns consistently exhibit

heavier tails than the Gaussian distribution predicts [12]. The consequences are severe: a Gaussian model that underestimates tail risk leads to insufficient capital buffers, poorly hedged positions, and failure to detect clustering of losses during market stress.

This work investigates, through systematic Monte Carlo experimentation, how VaR and ES estimators behave when the data generating process departs from normality. Specifically, we study:

- the large sample distribution of quantile estimators and its breakdown in sparse tail regions, studied empirically via bootstrap resampling from historical returns;
- finite sample concentration guarantees for the empirical CDF via the DKW inequality, compared against asymptotic plug in confidence intervals;
- variance reduction strategies, antithetic variates and portfolio direction mean shift importance sampling, and their effectiveness depends critically on the choice of proposal distribution. While simple mean-shift strategies provide modest improvements, more sophisticated rare-event simulation techniques may yield substantially greater efficiency gains for tail estimation;
- the practical consequences of model misspecification, where synthetic return paths are generated from a Student t distribution and forecast with a Gaussian model;
- time varying volatility via GARCH(1,1) models on portfolio returns, with particular attention to the improvement from Student t innovations.

These questions connect directly to the challenge of robust decision making under distributional uncertainty. Estimation error in tail quantities translates directly into unreliable risk based decisions. Understanding the sources and magnitude of this error is a necessary precursor to building more robust frameworks.

2 Problem Setup

2.1 Portfolio Returns and Losses

Let R denote the one period return of a portfolio and define the corresponding loss as

$$L = -R.$$

Fix confidence level $\alpha \in (0, 1)$, typically $\alpha = 0.99$ for regulatory applications. Portfolio loss is formed as

$$L = -\boldsymbol{w}^\top \boldsymbol{X},$$

where $\boldsymbol{w}$ is a fixed weight vector and $\boldsymbol{X}$ is the vector of asset returns. Weights are normalized to sum to unity.

2.2 Gaussian Baseline

Under the parametric Gaussian model, asset returns follow

$$\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are estimated from historical data as the sample mean vector and sample covariance matrix with the standard unbiased correction. This model assigns exponentially small probabilities to extreme losses and serves as the baseline for comparison.

2.3 Heavy Tailed Extension: Multivariate Student t

To capture tail thickness, we employ the multivariate Student t distribution via its scale mixture representation[13]:

$$\boldsymbol{Z} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}), \qquad U \sim \chi^2_\nu \quad \text{(independent of } \boldsymbol{Z}\text{)}$$

$$\boldsymbol{X} = \boldsymbol{\mu} + \frac{\boldsymbol{Z}}{\sqrt{U/\nu}}.$$

This yields

$$\boldsymbol{X} \sim t_\nu(\boldsymbol{\mu}, \boldsymbol{\Sigma}).$$

The degrees of freedom parameter ν governs tail thickness: as $\nu \to \infty$, the distribution converges to Gaussian; for small ν, tails are substantially heavier. The construction requires $\nu > 2$ to guarantee finite covariance. In the baseline heavy tail experiment, $\nu = 5$ is fixed. For the misspecification experiment, the Student t scale matrix is set as

$$\boldsymbol{\Sigma}_{\text{true}} = \boldsymbol{\Sigma}_{\text{hist}} \frac{\nu_{\text{true}} - 2}{\nu_{\text{true}}},$$

ensuring the model covariance matches the historical sample covariance at the true degrees of freedom.

2.4 Risk Measure Definitions

For a loss random variable L with CDF F:

$$\text{VaR}_\alpha(L) = \inf\{\ell : \Pr(L \le \ell) \ge \alpha\} = F^{-1}(\alpha),$$

$$\text{ES}_\alpha(L) = \mathbb{E}[L \mid L \ge \text{VaR}_\alpha(L)].$$

VaR is the α quantile of the loss distribution. ES is the expected loss conditional on exceeding VaR. ES satisfies the coherence axioms of Artzner et al. (1999)[1], including subadditivity, monotonicity, positive homogeneity, and translation invariance, whereas VaR does not satisfy subadditivity in

general, although it does under specific distributional assumptions such as elliptical distributions (including the Gaussian case). Both are estimated empirically: VaR as the sample α quantile using linear interpolation, and ES as the mean of realized losses exceeding the VaR estimate.

2.5 Why Estimation is Difficult under Heavy Tails

At confidence level $\alpha = 0.99$ with a rolling estimation window of $n = 500$ observations, the expected number of tail observations is approximately $n(1-\alpha) = 5$. Under heavy tails with finite but large variance, for example $\nu = 5$, these observations are highly dispersed. The asymptotic variance of the sample quantile is proportional to $1/f(q_\alpha)^2$, the reciprocal of the squared density at the quantile [3, 4]. Heavy tailed distributions have smaller $f(q_\alpha)$ than Gaussian distributions at the same probability level, inflating estimator variance. A Gaussian kernel density estimate of this density using Silverman's bandwidth

$$h = 0.9\,\hat{\sigma}\,n^{-1/5}$$

becomes unreliable when the effective tail sample is small.

3 Methodology

3.1 Monte Carlo Scenario Generation

Monte Carlo methodology follows [8]. Two scenario generation methods are implemented. The parametric Gaussian Monte Carlo draws 20,000 i.i.d. scenarios from the fitted multivariate Gaussian distribution using a seeded pseudo random number generator. The historical bootstrap resamples return vectors with replacement from the historical record, preserving empirical dependence structure. Both methods use $\alpha = 0.99$.

Portfolio losses are

$$L^{(i)} = -\boldsymbol{w}^\top \boldsymbol{X}^{(i)}.$$

VaR is the empirical α quantile of the simulated loss distribution using linear interpolation, and ES is the mean of losses exceeding the VaR estimate.

3.2 Asymptotic Theory for Quantile Estimators

The quantile CLT is standard [3, 4]. To study finite sample behavior empirically, the full historical portfolio return series is treated as a proxy population and resampled with replacement for sample sizes $n \in \{250, 500, 1000, 2000\}$ over 200 independent repetitions. A high precision reference VaR is computed from 200,000 bootstrap draws from the same historical series. Under regularity

conditions, the quantile central limit theorem gives

$$\sqrt{n}\,(\hat{q}_\alpha - q_\alpha) \;\Rightarrow\; \mathcal{N}\!\left(0,\, \frac{\alpha(1-\alpha)}{f(q_\alpha)^2}\right).$$

The plug in confidence interval substitutes a Gaussian kernel density estimate $\hat{f}(\hat{q}_\alpha)$ evaluated at the sample quantile, using Silverman's bandwidth. While kernel density estimation is used here as a plug-in method for asymptotic inference, its performance in extreme tail regions is known to be unstable due to data sparsity. Consequently, results relying on $\hat{f}(q_\alpha)$ should be interpreted with caution. The standard error is

$$\mathrm{se} = \sqrt{\frac{\alpha(1-\alpha)}{n\,\hat{f}^2}},$$

and the critical value is obtained from a rational approximation to the inverse normal CDF. RMSE is measured relative to the bootstrap reference VaR, and empirical coverage is the fraction of repetitions in which the reference falls within the confidence interval.

3.3 Finite Sample Bounds via the DKW Inequality

The Dvoretzky Kiefer Wolfowitz inequality [5, 6] provides a distribution free, non asymptotic bound on the empirical CDF:

$$\Pr\!\left(\sup_x |F_n(x) - F(x)| > \varepsilon\right) \le 2\exp(-2n\varepsilon^2).$$

For confidence level $1-\delta$ with $\delta = 0.05$ to match the asymptotic CI, the calibrated bandwidth is

$$\varepsilon = \sqrt{\frac{\log(2/\delta)}{2n}}.$$

The quantile bracket is

$$q_{\alpha-\varepsilon}(F_n) \le q_\alpha(F) \le q_{\alpha+\varepsilon}(F_n),$$

with probabilities clipped to $[0, 1]$. This bracket is valid for all sample sizes $n \ge 1$ and any distribution with a well-defined cumulative distribution function. The same sample size grid and repetition structure as Section 3.2 is used, enabling a direct comparison of coverage and interval width between asymptotic and finite sample methods.

3.4 Variance Reduction

Variance reduction techniques follow [8]. Antithetic variates pair each Gaussian draw $\boldsymbol{Z}$ with its negation $-\boldsymbol{Z}$ in the multivariate normal space, then shift both by the mean $\boldsymbol{\mu}$. Standard deviation and RMSE of the VaR and ES estimators are evaluated over 200 independent repetitions of 20,000 simulations each, relative to a plain MC reference computed from 300,000 draws to minimize

reference error.

Importance sampling via mean shift shifts the sampling distribution toward the loss tail. For portfolio volatility

$$\sigma_p = \sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w}}$$

and shift scale k, the proposal mean is displaced by

$$\boldsymbol{\theta} = -(k\sigma_p)\,\frac{\boldsymbol{\Sigma}\boldsymbol{w}}{\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w}}.$$

This construction guarantees

$$\boldsymbol{w}^\top \boldsymbol{\theta} = -k\sigma_p,$$

a controlled shift of exactly k portfolio standard deviations toward losses. Likelihood ratio weights

$$w(x) = \frac{f_0(x)}{f_1(x)}$$

are stabilized by subtracting the log weight maximum before exponentiation. The effective sample size diagnostic is

$$\mathrm{ESS} = \frac{1}{\sum_i \tilde{w}_i^2}.$$

Shift scales $k \in \{0.5, 1.0, 1.5, 2.0\}$ are studied to characterize the variance reduction trade off.

3.5 Model Misspecification Analysis

Student t modeling follows [12]. A synthetic return path of length $T = 2{,}500$ is generated from a multivariate Student t with $\nu_{\text{true}} = 5$ and scale matrix set to match the historical sample covariance. Rolling VaR forecasts are then produced under two competing models over a window of $n = 500$: (A) a misspecified Gaussian model, approximated by setting $\nu_{\text{model}} = 1000$ and hence effectively Gaussian, and (B) a correctly specified Student t model with $\nu_{\text{model}} = 5$. The Kupiec unconditional coverage test quantifies miscoverage under each assumption.

3.6 Dynamic Volatility: Univariate GARCH(1,1)

GARCH models follow [9, 10]. Portfolio returns are obtained by projecting the asset return matrix onto the weight vector, yielding a scalar return series. A univariate GARCH(1,1) is then fit to this series. This approach avoids the computational cost and parameter proliferation of multivariate GARCH, while capturing the dominant source of return heteroskedasticity. For numerical stability during optimization, returns are scaled by a factor of 100 before fitting and rescaled afterward. The model dynamics are

$$r_t = \mu + \sigma_t z_t, \qquad \sigma_t^2 = \omega + \alpha_g\,\varepsilon_{t-1}^2 + \beta_g\,\sigma_{t-1}^2.$$

For the Gaussian variant, Section 4.5, GARCH Normal, innovations follow $z_t \sim \mathcal{N}(0,1)$. For the Student t variant, GARCH t, $z_t \sim t_\nu$ with ν estimated jointly by maximum likelihood. The one step ahead VaR forecast in original return units is

$$\mathrm{VaR}_{t+1} = -\frac{\hat{\mu} - z_\alpha \hat{\sigma}}{100},$$

where $z_\alpha = \Phi^{-1}(\alpha)$ under Gaussian innovations or $z_\alpha = t_\nu^{-1}(\alpha)$ under Student t innovations. To reduce computational cost, the model is refitted every 20 observations rather than daily, with the most recent fit used for intermediate forecasts.

3.7 Backtesting and Proper Scoring

Backtesting follows [14, 15]. The rolling backtest covers $n = 1{,}547$ test observations with a 500 day estimation window. Exception indicators are defined as

$$I_t = 1\{L_t > \mathrm{VaR}_{\alpha,t}\},$$

with nominal exception probability $p = 1 - \alpha = 0.01$.

The Kupiec unconditional coverage (UC) test compares the observed exception count k in n observations against the binomial null via a likelihood ratio statistic asymptotically distributed as $\chi^2(1)$:

$$LR_{\mathrm{UC}} = -2\left[(n-k)\log\left(\frac{1-p}{1-\hat{p}}\right) + k\log\left(\frac{p}{\hat{p}}\right)\right], \qquad \hat{p} = \frac{k}{n}.$$

The Christoffersen independence test fits a two state Markov chain to the exception sequence and tests whether exceptions are independently distributed over time, using a $\chi^2(1)$ LR statistic.

The Fissler–Ziegel scoring rule [16, 17] provides a strictly consistent loss function for joint VaR ES evaluation:

$$S_t = \frac{1}{E_t}\left((\mathbf{1}\{L_t \le V_t\} - \alpha)(V_t - L_t) + (1-\alpha)(E_t - V_t)\right) + \log(E_t),$$

with $E_t > 0$. Lower mean FZ0 scores indicate better joint calibration. The ES ratio is defined as the mean realized loss on exception days divided by the mean forecasted ES over all test days; values near 1 provide a heuristic indication of alignment between realized tail losses and forecasted ES, although this metric does not constitute a formal statistical test.

4 Experiments and Results

4.1 Gaussian vs. Student t Tail Behavior

The parametric Gaussian and historical bootstrap scenarios each use 20,000 draws at $\alpha = 0.99$. The heavy tail comparison fixes $\nu = 5$ for a controlled comparison rather than fitting ν from data. For consistency, the multivariate Student-t model is parameterized such that its covariance matches the empirical covariance matrix. Specifically, if

$$X \sim t_\nu(\mu, \Sigma),$$

the scale matrix is set to

$$\Sigma = \frac{\nu - 2}{\nu} \Sigma_{\text{hist}},$$

so that

$$\text{Cov}(X) = \Sigma_{\text{hist}}.$$

This ensures that differences between the Gaussian and Student-t models reflect tail behavior rather than scale discrepancies. Both VaR and ES increase substantially under the Student t assumption, illustrating the systematic tail underestimation inherent in Gaussian models applied to heavy tailed data.

4.2 Quantile Asymptotics and Finite Sample Bounds

RMSE of the sample quantile decays approximately as $n^{-1/2}$ on a log log scale, consistent with the CLT prediction. However, empirical coverage of the plug in confidence interval falls below the nominal 0.95 level at small n, because the kernel density estimate $\hat{f}(\hat{q}_\alpha)$ is unreliable in the sparse tail region. When this estimate approaches zero, the confidence interval width inflates to the point of being uninformative.

The DKW bracket achieves nominal or super nominal empirical coverage across all sample sizes, confirming the finite sample guarantee. The width comparison reveals the conservatism cost: the DKW bracket is substantially wider than the asymptotic interval at large n. The two methods are evaluated at the same 95% confidence level, enabling a direct width versus coverage comparison.

4.3 Variance Reduction

Antithetic variates reduce the standard deviation of the VaR estimator modestly relative to plain Monte Carlo. The reduction is limited because the VaR quantile functional is non smooth and does not admit a first order Taylor expansion that the antithetic pairing can exploit. Gains are somewhat larger for ES, which involves a tail average and is smoother.

Mean shift importance sampling at $k = 1.0$ one portfolio standard deviation toward losses reduces both standard deviation and RMSE of the VaR estimator. At $k = 2.0$, weight instability emerges: a small fraction of simulated scenarios receive very large likelihood ratio weights, inflating variance. The ESS diagnostic drops significantly at large shift scales, signaling effective sample size collapse and illustrating the fundamental tension between concentrating probability mass in the tail and maintaining stable importance weights.

4.4 Model Misspecification

When the true DGP is Student t with $\nu = 5$ and VaR is forecast with the misspecified Gaussian model, the exception rate over the $T = 2{,}500$ observation synthetic path substantially exceeds the nominal $p = 0.01$. The Kupiec LR statistic is large and the p value is small, rejecting correct coverage. Under the correctly specified Student t model with $\nu = 5$, exception rates converge toward the nominal level and the test fails to reject.

This experiment isolates distributional misspecification from other sources of error: the sample size, window, and portfolio composition are identical across models, and only the tail assumption differs. The result confirms that distributional misspecification alone is sufficient to produce systematically unreliable VaR forecasts at the 99% level.

4.5 Rolling Backtests on Real Returns

Table 1 presents the central empirical finding: all four models are backtested on the same real return series with $n = 1{,}547$ test observations and a 500 day rolling estimation window.

Table 1: Rolling 99% VaR and ES Backtest Results ($n = 1{,}547$ test observations, window = 500 days)

Model	Exception Rate	Kupiec p UC	Christoffersen p IND	ES Ratio	FZ0 Mean
Normal (Gaussian)	0.02198	4.35×10^{-5}	0.0417	1.122	-3.210
Historical Bootstrap	0.01681	0.0142	0.0083	0.870	-2.943
GARCH(1,1) Normal	0.02262	1.85×10^{-5}	0.2409	0.925	-3.390
GARCH(1,1) t	0.00905	0.703	0.613	0.759	-3.078

Bold row, GARCH(1,1) t, indicates best calibrated model. Nominal exception rate is 0.01. ES ratio equals mean realized loss on exception days divided by mean forecasted ES; values near 1 provide a heuristic indication of alignment between realized tail losses and forecasted ES, although this metric does not constitute a formal statistical test. FZ0 mean: lower is better. n exceed: count of VaR breaches.

The static Gaussian model produces an exception rate of 2.20%, more than twice the nominal 1%, with the Kupiec test rejecting decisively, p = 4.35×10^{-5}. The Christoffersen test also rejects, p

= 0.042, indicating exception clustering: breaches concentrate during periods of elevated volatility. The ES ratio of 1.122 shows that realized losses on exception days exceed the model's ES forecast on average, confirming tail underestimation.

The historical bootstrap achieves a lower exception rate, 1.68%, and a marginally passing Kupiec test, p = 0.014, but the Christoffersen test rejects more strongly, p = 0.008. The nonparametric approach partially captures tail thickness from the historical sample but does not adapt to time varying volatility, causing exceptions to cluster.

GARCH(1,1) Normal resolves the clustering problem: the Christoffersen test passes comfortably, p = 0.241, confirming that conditioning on recent volatility removes temporal dependence in exceptions. However, the Kupiec test still rejects, p = 1.85×10^{-5}: the model produces 35 exceptions against a nominal expectation of approximately 15.5. Conditional volatility adaptation is necessary but not sufficient, the Gaussian innovation assumption still underestimates tail probability.

GARCH(1,1) t achieves the best calibration across all diagnostics: exception rate 0.905% (14 breaches versus nominal 15.5), Kupiec p = 0.703, Christoffersen p = 0.613. Both tests fail to reject. The ES ratio of 0.759 indicates mild conservatism. The FZ0 mean of -3.078 is intermediate; notably, GARCH(1,1) Normal achieves the best FZ0 score, -3.390, despite failing both backtests, illustrating that proper scoring and calibration tests are complementary diagnostics.

5 Discussion

5.1 Instability of Tail Estimation

The observed undercoverage of asymptotic confidence intervals is not solely attributable to kernel density estimation error, but more fundamentally reflects slow convergence in extreme quantiles. At high confidence levels, the effective number of observations in the tail is small, and even consistent estimators of the density $f(q_\alpha)$ exhibit high variance. As a result, standard error estimates based on

$$\mathrm{Var}(\hat{q}_\alpha) \approx \frac{\alpha(1-\alpha)}{nf(q_\alpha)^2}$$

become unstable in finite samples, leading to degraded coverage accuracy of asymptotic intervals.

The variance reduction experiments reveal a related constraint: methods that reduce variance for smooth functionals provide limited or unstable gains for non smooth quantile estimators. Importance sampling is most effective at moderate shift scales; beyond one or two portfolio standard deviations of shift, weight collapse reduces the effective sample size and can increase variance rather than reduce it.

An alternative framework for tail risk estimation is provided by Extreme Value Theory (EVT), particularly the peaks-over-threshold approach [18], which directly models tail behavior beyond high thresholds. While not explored in this study, EVT provides a complementary methodology

for extreme quantile estimation.

5.2 Implications for Decision Making

From a decision theoretic perspective, the empirical results have direct operational implications. A risk manager relying on static Gaussian VaR estimates at the 99% level should expect breaches at approximately twice the nominal rate, and those breaches will cluster during stress periods, precisely when reliable risk estimates are most critical.

The GARCH experiments demonstrate that two distinct improvements are needed simultaneously: dynamic volatility adaptation to address exception clustering, and distributional tail flexibility to achieve nominal unconditional coverage. Neither alone is sufficient. GARCH Normal achieves the former but not the latter; GARCH t achieves both.

The Fissler Ziegel scoring rule provides a complementary diagnostic to the binary pass fail of backtests. A model can score well on a proper loss function without achieving nominal unconditional coverage. Using multiple diagnostics, the UC test, independence test, ES ratio, and proper scoring, gives a more complete picture of model reliability.

6 Conclusion

This study examines the behavior of Monte Carlo estimators for Value at Risk (VaR) and Expected Shortfall (ES) under heavy-tailed return distributions, model misspecification, and time-varying volatility.

The results indicate that both distributional assumptions and volatility dynamics play a central role in determining the reliability of tail risk estimates. Models incorporating time-varying volatility and heavy-tailed innovations, such as GARCH(1,1) with Student-t errors, demonstrate improved calibration relative to static Gaussian benchmarks.

At the same time, finite-sample limitations remain substantial. In extreme quantile estimation, the effective sample size in the tail is inherently small, leading to high estimator variance and instability in inference procedures. This affects both asymptotic confidence intervals and simulation-based estimators.

Overall, the findings highlight the importance of combining flexible distributional modeling with robust inference techniques. Future work may benefit from integrating extreme value methods and advanced rare-event simulation approaches to further improve tail risk estimation.

Reproducibility Note

Reproducibility Note

All experiments are implemented in Python. Equity return data are obtained via the `yfinance` API for a fixed set of large-cap assets (AAPL, MSFT, AMZN, NVDA, JPM, TLT), using daily adjusted closing prices from January 2018 onward. Returns are computed as simple percentage changes and aligned across assets.

Experiments follow a modular design: each study is defined by a set of parameters (sample size, degrees of freedom, shift scale, estimation window, and confidence level) and executed through a unified pipeline. Outputs include summary statistics in tabular form and diagnostic figures.

The core statistical components are multivariate normal and Student-t simulation, asymptotic quantile confidence intervals via Silverman bandwidth kernel density estimation, DKW finite-sample quantile brackets, antithetic and importance sampling variance reduction, Kupiec and Christoffersen backtests, univariate GARCH(1,1) with Gaussian and Student-t innovations fit by maximum likelihood, and the Fissler–Ziegel FZ0 scoring rule. Random seeds are fixed per experiment to ensure reproducibility conditional on the dataset.

Disclaimer

This work is a methodological and computational study of statistical estimation under heavy tailed distributions. It does not constitute financial advice, investment recommendations, or guidance on regulatory compliance. All empirical results are presented solely for scientific and educational purposes. The authors make no representations regarding the suitability of any model or method described herein for any specific financial application.

References

[1] Artzner, P., Delbaen, F., Eber, J.-M., Heath, D. (1999). Coherent Measures of Risk. *Mathematical Finance*.

[2] McNeil, A., Frey, R., Embrechts, P. (2005). *Quantitative Risk Management*. Princeton University Press.

[3] van der Vaart, A. (1998). *Asymptotic Statistics*. Cambridge University Press.

[4] Serfling, R. (1980). *Approximation Theorems of Mathematical Statistics*. Wiley.

[5] Dvoretzky, A., Kiefer, J., Wolfowitz, J. (1956). Asymptotic Minimax Character of the Sample Distribution Function. *Annals of Mathematical Statistics*.

[6] Massart, P. (1990). The Tight Constant in the DKW Inequality. *Annals of Probability*.

[7] Silverman, B. (1986). *Density Estimation for Statistics and Data Analysis*. Chapman and Hall.

[8] Glasserman, P. (2004). *Monte Carlo Methods in Financial Engineering*. Springer.

[9] Engle, R. (1982). Autoregressive Conditional Heteroskedasticity. *Econometrica*.

[10] Bollerslev, T. (1986). Generalized ARCH. *Journal of Econometrics*.

[11] Bollerslev, T. (1987). A Conditional Heteroskedastic Time Series Model for Speculative Prices. *Review of Economics and Statistics*.

[12] Cont, R. (2001). Empirical Properties of Asset Returns. *Quantitative Finance*.

[13] Kotz, S., Nadarajah, S. (2004). *Multivariate t Distributions*. Cambridge University Press.

[14] Kupiec, P. (1995). Techniques for Verifying the Accuracy of Risk Measurement Models. *Journal of Derivatives*.

[15] Christoffersen, P. (1998). Evaluating Interval Forecasts. *International Economic Review*.

[16] Fissler, T., Ziegel, J. (2016). Higher Order Elicitability and Osband's Principle. *Annals of Statistics*.

[17] Gneiting, T., Raftery, A. (2007). Strictly Proper Scoring Rules. *Journal of the American Statistical Association*.

[18] Embrechts, P., Klüppelberg, C., Mikosch, T. (1997). *Modelling Extremal Events*. Springer.